

Statement of Work (SOW)

AWS-Powered IoT Platform for Argo Glasses Ecosystem

Prepared By: Kaynes Technology Limited

Document Version: 1.0

Document Type: Statement of Work (SOW)

1. Executive Summary

As the Argo Glasses platform scales across users, devices, and deployments, the need for centralized device management, operational visibility, secure media storage, monitoring, and analytics becomes increasingly important.

While the Android and iOS SDKs provide direct BLE and Wi-Fi communication with Argo Glasses devices, a cloud-native backend platform is required to support device lifecycle management, telemetry processing, media management, security controls, observability, and future platform expansion.

Kaynes Technology Limited proposes the implementation of an AWS-based IoT platform that complements the existing SDK ecosystem and establishes a secure, scalable, and enterprise-ready cloud foundation for the Argo Glasses solution.

2. Business Challenges

The current architecture primarily relies on direct communication between mobile applications and Argo Glasses devices.

As deployments grow, the following challenges emerge:

- Lack of centralized device visibility.
- No centralized telemetry repository.
- Difficulty monitoring battery health and device status.
- Limited media storage and archival capabilities.
- Lack of operational dashboards and alerting.
- Challenges in securing device communications at scale.
- Future requirements for OTA firmware updates.



- Need for enterprise-grade observability and diagnostics.
-

3. Project Objectives

The proposed AWS solution will:

- Enable centralized device management.
 - Collect and process device telemetry.
 - Provide secure cloud APIs.
 - Store and manage captured media.
 - Enable monitoring and operational visibility.
 - Support future OTA readiness.
 - Improve scalability and security.
 - Establish a cloud foundation for future analytics initiatives.
-

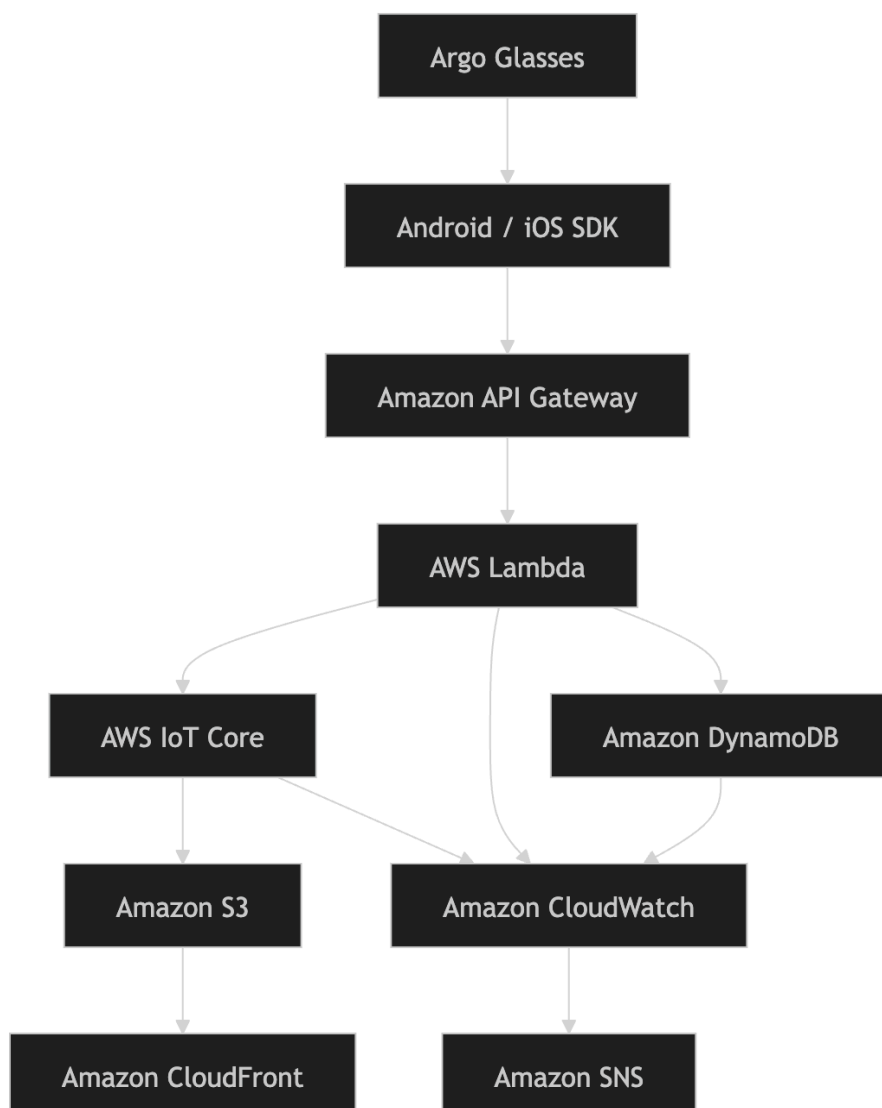
4. Existing Scope Understanding

The Argo Glasses SDK currently supports:

- Device discovery via BLE.
- Device connection management.
- Image capture.
- Audio recording control.
- Video recording control.
- Volume management.
- Battery status monitoring.
- Wi-Fi media transfer.
- Image download.
- Image deletion.
- Device status monitoring.
- Local live view.
- Error handling and diagnostics.

The AWS platform will complement these capabilities without replacing existing BLE or Wi-Fi workflows.

5. Proposed AWS Architecture



Architecture Components

AWS IoT Core

- Device registration
- Telemetry ingestion
- Device state management
- Device communication services

AWS IoT Device Management

- Fleet organization
- Device grouping
- Device lifecycle management
- Future OTA readiness

Amazon API Gateway

- Secure backend APIs
- Mobile application integration
- Rate limiting and throttling

AWS Lambda

- Event processing
- Business logic execution
- Telemetry transformation

Amazon DynamoDB

- Device metadata storage
- Telemetry indexing
- User-device relationships

Amazon S3

- Image storage
- Video archival
- Media repository

Amazon CloudFront

- Accelerated media delivery
- Secure content distribution



Amazon Cognito

- User authentication
- Identity management
- Mobile application authorization

Amazon CloudWatch

- Operational monitoring
- Log aggregation
- Metrics collection

Amazon SNS

- Alerts and notifications
- Device health notifications

AWS KMS

- Encryption key management

AWS Secrets Manager

- Secure credential storage

6. AWS Service Mapping

Business Requirement	AWS Service
• Device Telemetry	• AWS IoT Core
• Device Fleet Management	• AWS IoT Device Management
• API Access	• Amazon API Gateway
• Event Processing	• AWS Lambda
• Metadata Storage	• Amazon DynamoDB
• Media Storage	• Amazon S3
• Content Delivery	• Amazon CloudFront

Business Requirement	AWS Service
• User Authentication	• Amazon Cognito
• Monitoring	• Amazon CloudWatch
• Notifications	• Amazon SNS
• Encryption	• AWS KMS
• Secret Management	• AWS Secrets Manager

7. Scope of Work

Phase 1 – Discovery & Architecture

- Requirement assessment
- Solution architecture design
- AWS landing zone planning
- Security review

Phase 2 – AWS Infrastructure Setup

- AWS account configuration
- IAM setup
- Networking configuration
- Security baseline implementation

Phase 3 – IoT Platform Implementation

- AWS IoT Core setup
- Device registry creation
- Device policies
- Telemetry ingestion framework

Phase 4 – Backend Services

- API Gateway implementation
- Lambda development
- DynamoDB integration

Phase 5 – Media Management

- S3 storage implementation
- CloudFront configuration
- Media lifecycle policies

Phase 6 – Monitoring & Security

- CloudWatch dashboards
- SNS alerts
- KMS encryption
- Secrets Manager implementation

Phase 7 – Testing & Validation

- Functional testing
- Security validation
- Performance testing
- Documentation review

8. Deliverables

- AWS Solution Architecture Document
 - Infrastructure Deployment
 - AWS IoT Configuration
 - API Gateway Configuration
 - Lambda Functions
 - DynamoDB Schema
 - S3 Media Repository
 - Monitoring Dashboards
 - Security Configuration
 - Technical Documentation
 - Knowledge Transfer Session
-

9. Project Timeline

Phase	Duration
• Discovery & Design	• 3 Week
• Infrastructure Setup	• 3 Week
• IoT Platform Setup	• 4 Week
• Backend Development	• 5 Week
• Testing & Validation	• 2 Week
• Handover & KT	• 3 Week

Total Duration: 20 Weeks

10. AWS Cost Estimate

Assumptions

- 5,000-10,000 connected devices
- Continuous telemetry transmission
- Daily image uploads
- Mobile application API usage
- Operational monitoring enabled
- Long-term media retention

Estimated Monthly AWS Consumption

AWS Service	Monthly Cost (USD)
• AWS IoT Core	• \$880
• AWS IoT Device Management	• \$420
• Amazon API Gateway	• \$560



AWS Service	Monthly Cost (USD)
• AWS Lambda	• \$290
• Amazon DynamoDB	• \$750
• Amazon S3	• \$520
• Amazon CloudFront	• \$250
• Amazon Cognito	• \$220
• Amazon CloudWatch	• \$470
• Amazon SNS	• \$95
• AWS KMS	• \$20
• AWS Secrets Manager	• \$25
Total Estimated Monthly Cost	~\$4,500

AWS Pricing Calculator

Calculator Link:

<https://calculator.aws/#/estimate?id=6b063ac9048ab1add7abd92c1ee2ef11e70642f1>

Note: Actual AWS consumption charges will vary based on device count, telemetry volume, storage utilization, API requests, data transfer, and production usage patterns.

11. Assumptions

- Existing SDK development remains unchanged.
- BLE communication continues through mobile applications.
- Argo Glasses firmware supports required telemetry generation.
- Internet connectivity is available for cloud interactions.
- Customer provides required AWS account access.



12. Out of Scope

- Mobile application development.
 - SDK development.
 - Firmware development.
 - Hardware modifications.
 - AI/ML model development.
 - Video analytics.
 - Custom dashboard development.
 - Third-party integrations not identified during discovery.
-

13. Acceptance Criteria

- The solution will be considered accepted when:
 - AWS environment is deployed successfully.
 - IoT Core receives telemetry from devices.
 - APIs are operational.
 - Media uploads are stored in S3.
 - Monitoring dashboards are functional.
 - Security controls are implemented.
 - Documentation is delivered.
 - Knowledge transfer is completed.
-

14. Support & Maintenance

Kaynes Technology Limited will provide:

- 6 months post-go-live support.
 - Infrastructure issue resolution.
 - AWS configuration support.
 - Monitoring and alert assistance.
 - Minor platform enhancements.
 - Documentation updates.
-

15. Conclusion

The proposed AWS-based IoT platform establishes a scalable and secure cloud foundation for the Argo Glasses ecosystem. By integrating AWS IoT, storage, monitoring, security, and API services, the solution enables centralized device management, operational visibility, media management, and future platform expansion while remaining aligned with the existing Android and iOS SDK architecture.
